

Jan Procházka | PRO0387

Ostrava 2026

GUI pro aplikaci analyzující schéma databáze a SQL dotazy

Vedoucí

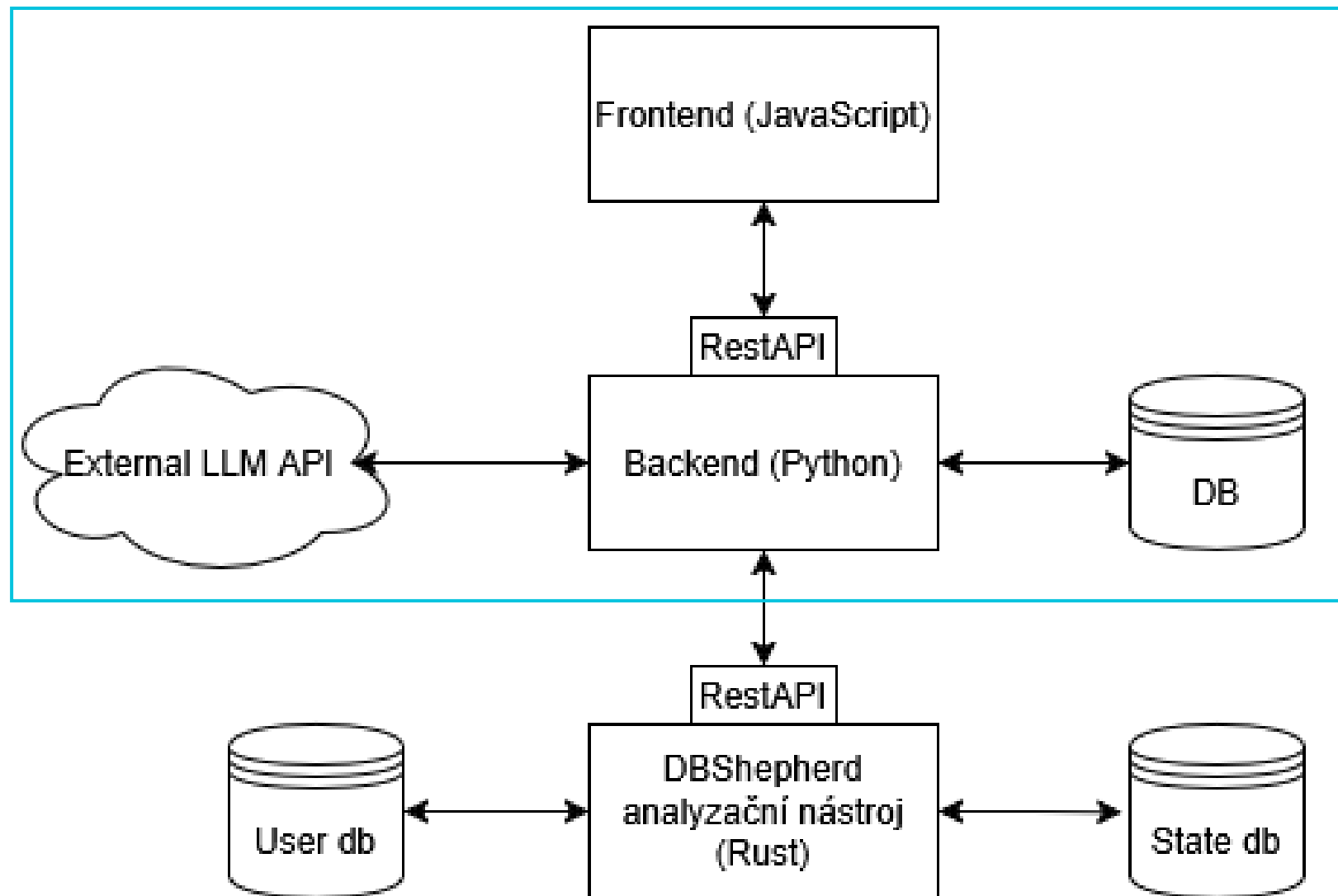
doc. Ing. Radim Bača, Ph.D.

Cíl práce

- DBShepherd – Nástroj pro analýzu databázových schémat a SQL dotazů
- Redundance
- Antipatterny
- CLI
- Webové rozhraní
- Komentář AI

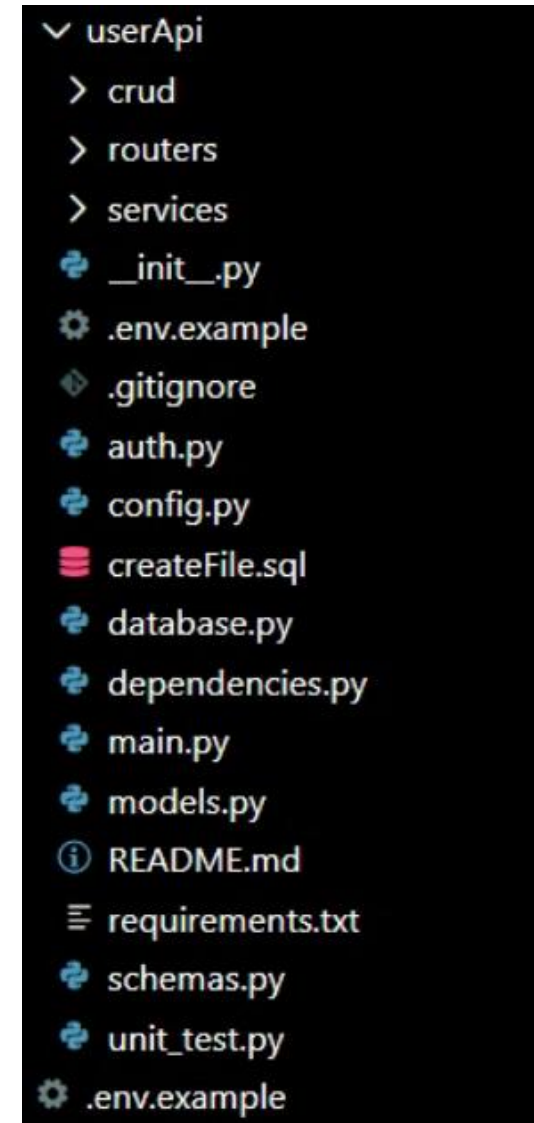
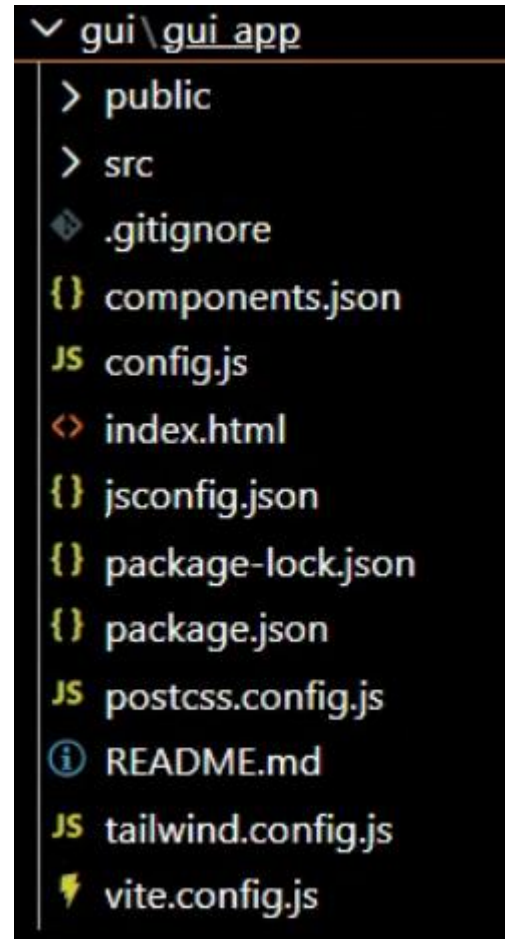
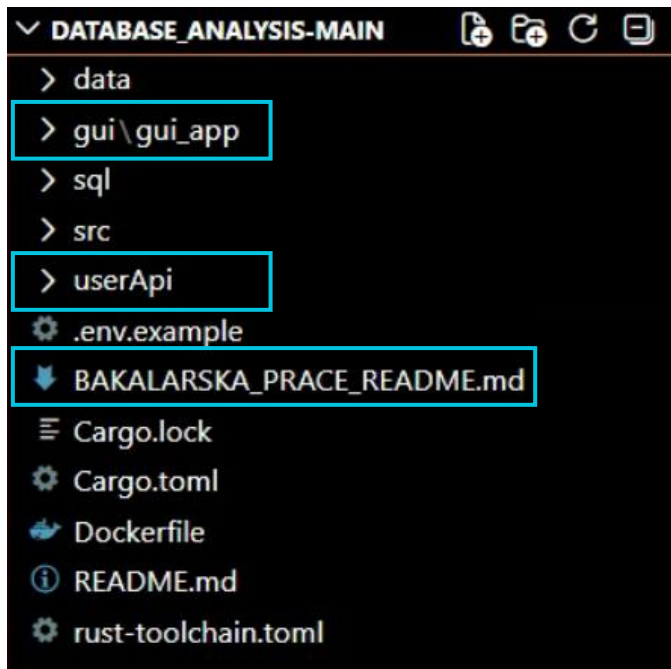
Architektura

- Backend
 - RestAPI
 - External LLM API
 - DB
- Frontend



Struktura projektu

- gui/
- userApi/
- BAKALRSKA_PRACE_README.md

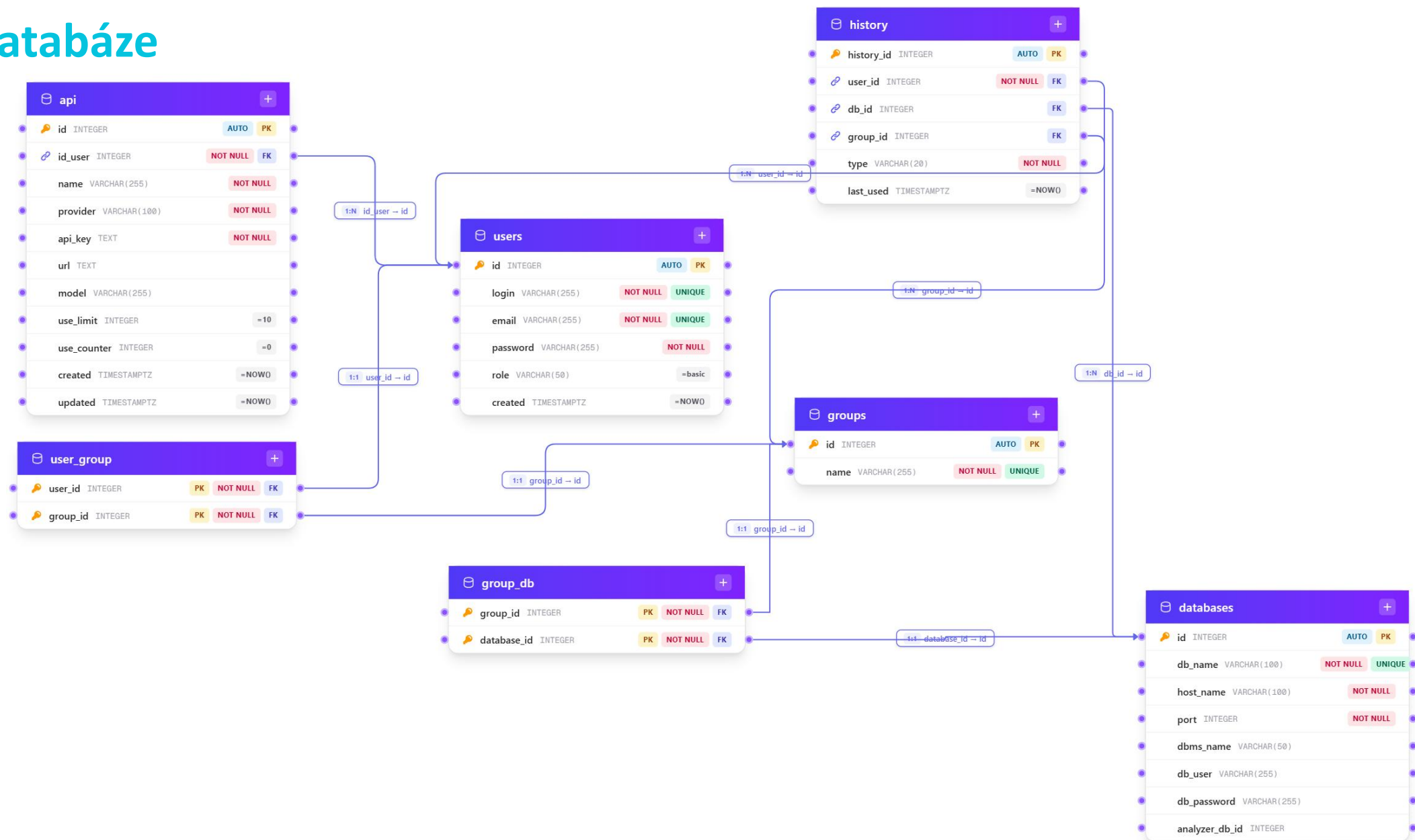


Použité technologie - Backend

- Python
- FastAPI
- bcrypt
- OAuth2PasswordBearer + JWT
- litellm

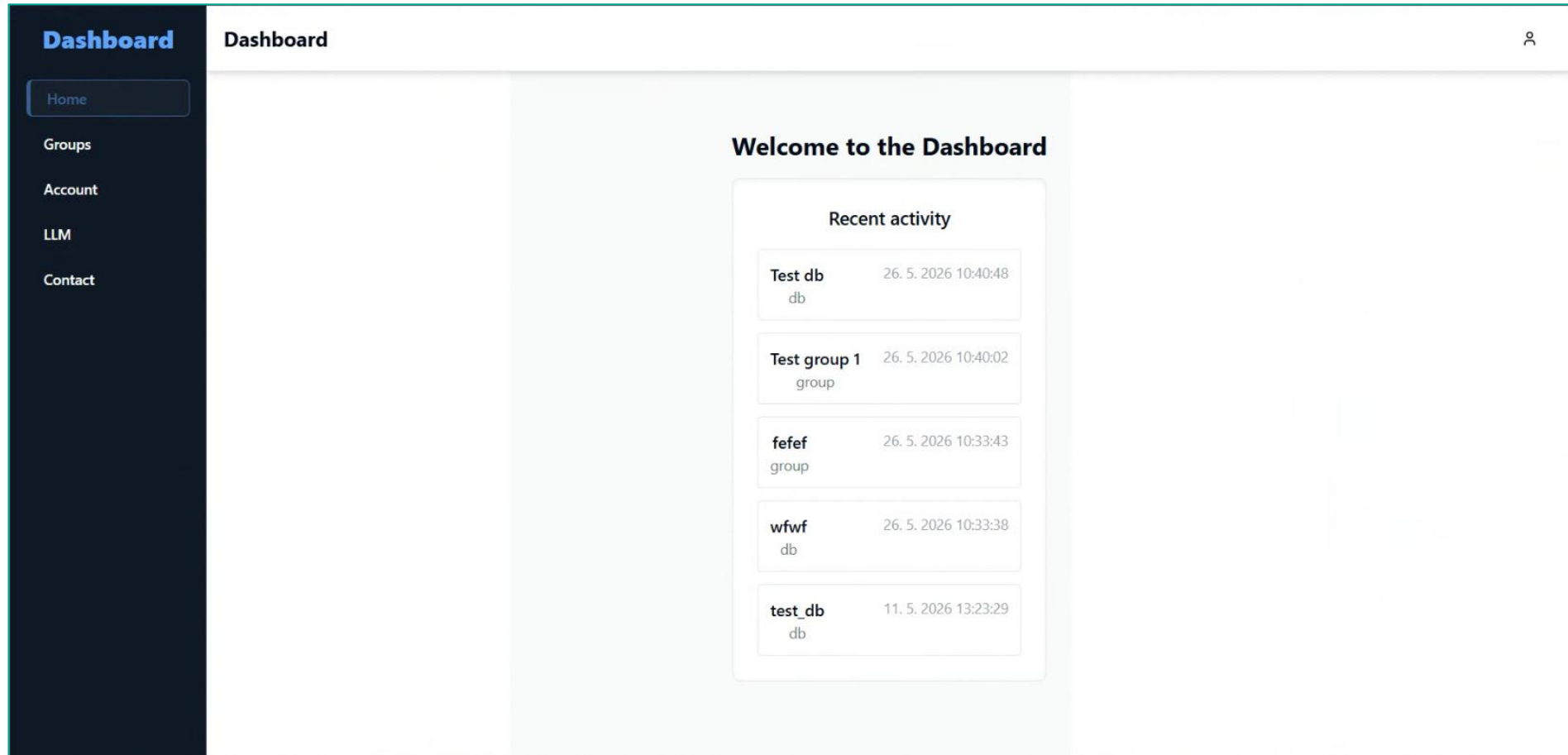


Databáze



Použité technologie — Frontend

- JavaScript
- React
- Vite
- shadcn/ui



Ukázka Frontendu

Dashboard

Home

Groups

Account

LLM

Contact

Test db

ChatGPT

Analyze

Rewrite Selected



SQL #1

Rewrite

```
SELECT id_person, (SELECT min(name) FROM person AS p2 WHERE p1.id_person = p2.id_person) FROM person AS p1
```

Errors:

- Consider removing this [Bare { table: "p1" }]

```
SELECT id_person, MIN(name) AS name FROM person GROUP BY id_person;
```



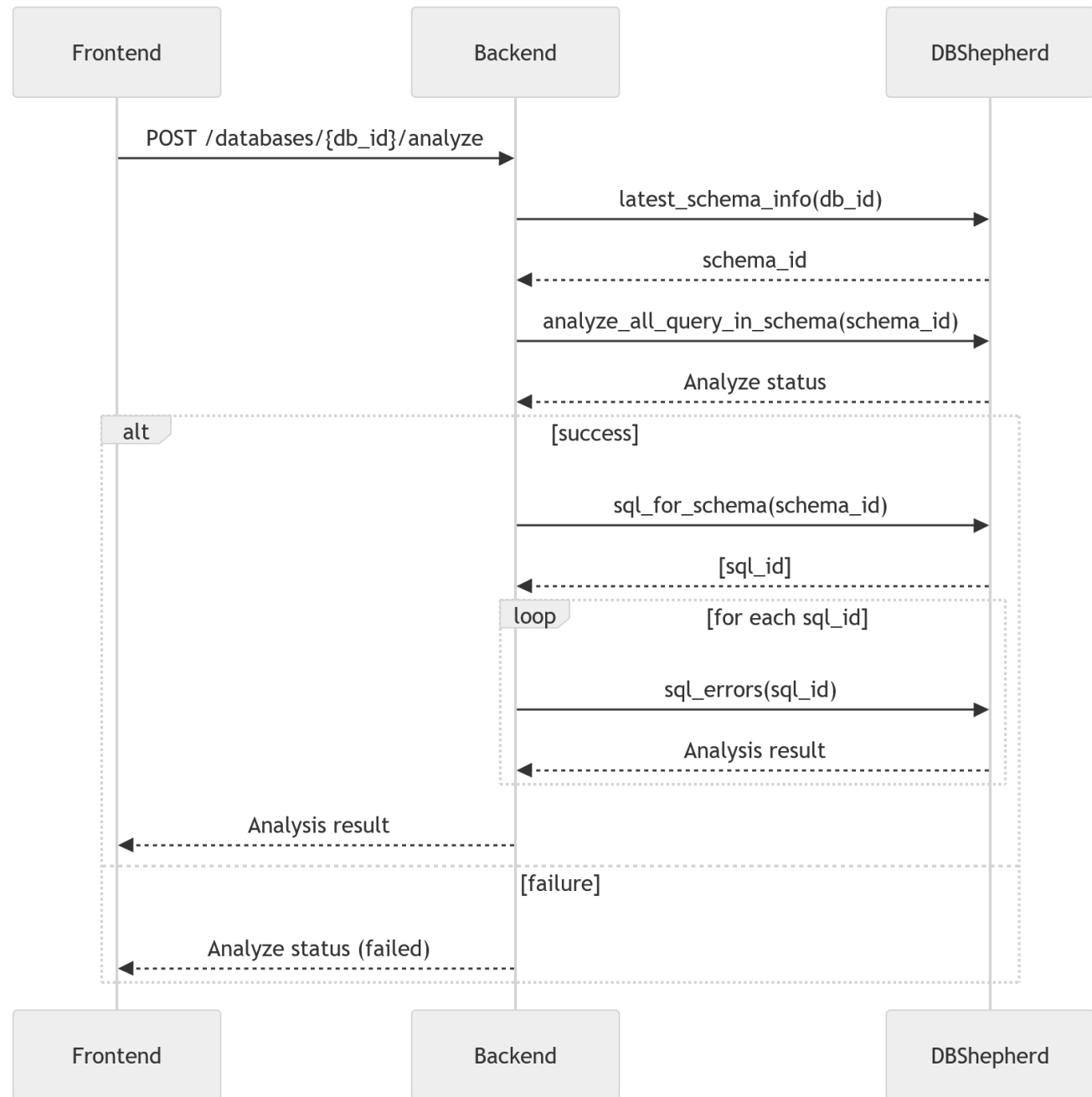
Enter SQL...

Insert

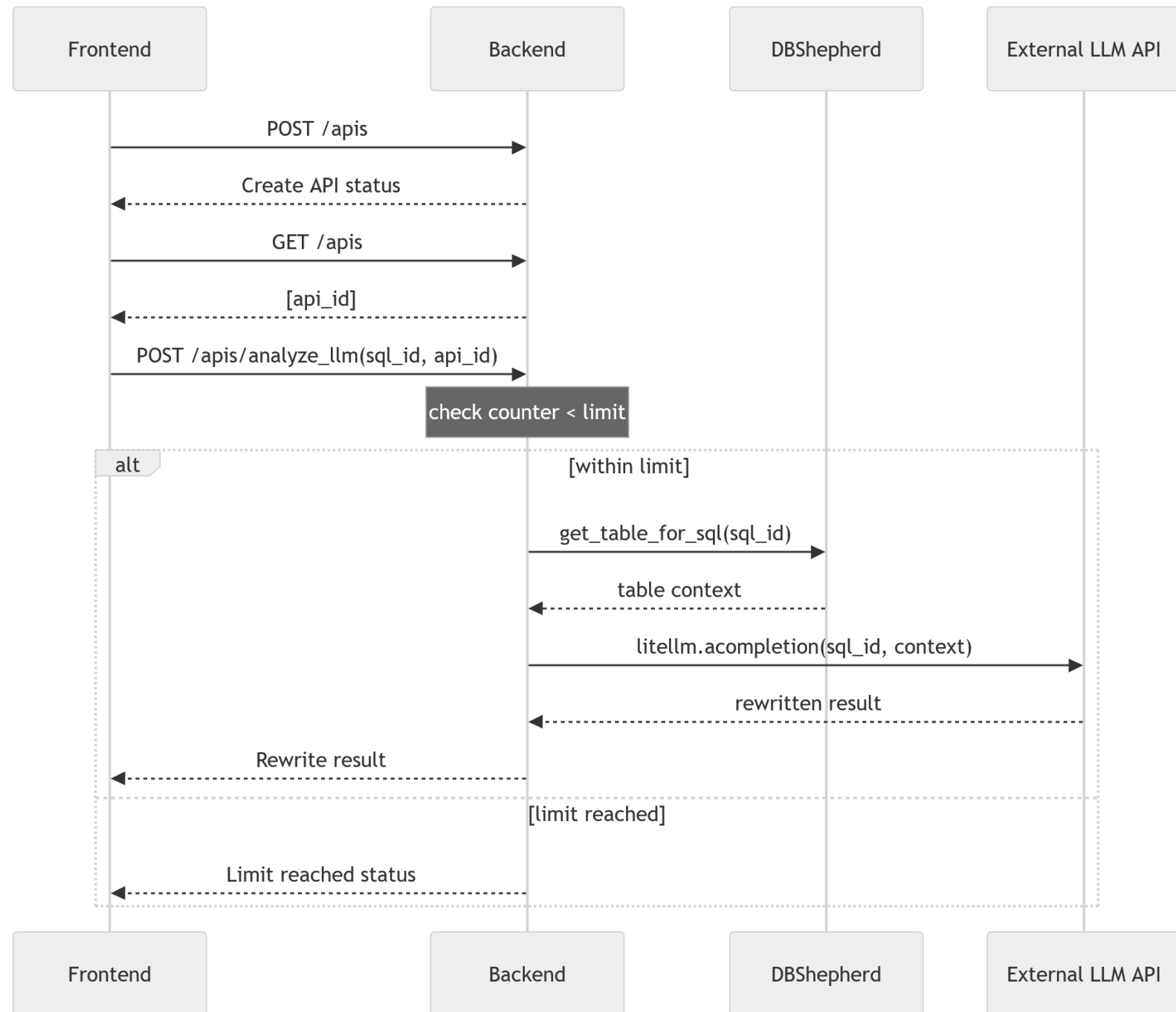
Enter DDL...

Insert DDL

Workflow analýzy



Workflow přepisu



Děkuji za pozornost

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Otázky

- Pokud máte nějaké otázky, rád je zodpovím